Form B

2. (a) H_0: age group and weekly church attendance are independent
H_a: age group and weekly church attendance are associated

(b) The smallest expected frequency is $\frac{(116)(117)}{523} = 25.95$, well beyond 5.
The requirements are met.

(c)

	Attend weekly	Not weekly
Observed	39	137
Expected	39.37	136.63

$\chi^2 = \frac{(39-39.37)^2}{39.37} + \frac{(137-136.63)^2}{136.63}$ + 4 other terms

30-49 contribution = $\underline{0.00448}$

(d) 1 - pchisq(2.6247, df = 2)

(e) We fail to reject that age group and weekly church attendance are independent

3. (a) gf_point(rent ~ value, data = rentalProperties)

(b) The residuals-vs-fitted plot (center graph) looks reasonably scattered about the zero line without a pattern, and the normal quantile plot is fairly straight (except at one end). The SLM seems (mostly) confirmed.

(c) The point estimate is $b_1 = 0.02471$ and $SE_{b_1} = 0.00346$
The 82% CI is

$$b_1 \pm t_c(SE_{b_1}) = 0.02471 \pm (1.3689)(0.00346), \text{ or } (0.0200, 0.0294)$$

(d) summary(lm(rent ~ value, data = rentalProperties))

(e) $\widehat{rent} = 5.2758 + 0.02471(value)$ (with both rent, value measured in the thousands)

(f) $\widehat{rent} = 5.2758 + 0.02471(280) = 12.195$,
or \$12,195.00.

(g) cor(rent ~ value, data = rentalProperties)
percent of variation explained = $R^2 = (0.7749)^2$, or $\underline{60.05\%}$

4. (a) A normal plot is a device to help determine if a quantitative variable seems to have a normal distribution.
It is unnecessary, here, as we were told in the problem description that the populations are normal.

(b) $\frac{S_{max}}{S_{min}} = \frac{1.5307}{0.6552} = 2.336 > 2$ The rule of thumb is <u>not satisfied</u>.

(c) $df1 = 4 - 1 = 3$

$df2 = (24 + 27 + 29 + 26) - 4 = 102$

$MSG = \frac{22.1273}{3} = 7.376$

$\frac{7.376}{MSE} = 6.2214 \Rightarrow MSE = 1.1856$

$\frac{SSE}{102} = 1.1856 \Rightarrow SSE = 120.93$

(d) 1 - pf(6.2214, df1 = 3, df2 = 102)

(e) Tukey pairwise comparisons are conducted only when the null hypothesis of ANOVA has already been rejected. Each line above is specifically about the difference of two group means — the first line is about $\mu_B - \mu_A$, the second about $\mu_C - \mu_A$, and so on. The significant p-values at $\alpha = 0.05$ appear on lines 1, 4 and 5: line 1 allows us to reject $H_0: \mu_B - \mu_A = 0$ in favor of $H_a: \mu_B - \mu_A \neq 0$.

Conclusions at the 5% level: $\left\{ \begin{array}{l} \mu_A \neq \mu_B \\ \mu_B \neq \mu_C \\ \mu_B \neq \mu_D \end{array} \right\}$ or μ_B is unequal to all other group means.

(f) TukeyHSD(len ~ grp, data = lengthByGrp)